README

The Ballistic N Type MOSFET - Equations and Parameters

The drain current approximated over the entire subthreshold, linear and saturation regions is equated using

$$\boxed{I_D \approx W \cdot |Q_n(V_{GS})| \cdot v_T \cdot \left(\frac{1 - e^{-qV_{DS}/k_BT}}{1 + e^{-qV_{DS}/k_BT}} \right)}$$

where,

$$Q_n(V_{GS}) = -C_{OX} \cdot m \cdot (k_BT/q) \cdot \ln\left(1 + e^{q(V_{GS} - V_T)/mk_BT}\right)$$

$$v_T = \sqrt{\frac{2k_BT}{\pi m^*}}$$

With the subthreshold effects included using

$$V_T = V_{T0} - \delta V_{DS}$$

Therefore, the parameters to be defined are

1. C_{OX}
2. V_{T0}
3. δ
4. m
5. T
6. W
7. $mr = m*/m_0$

Reference

[1] Essentials of MOSFETs - Mark Lundstrom